FIELD GUIDE — NETWORK CAPTURE

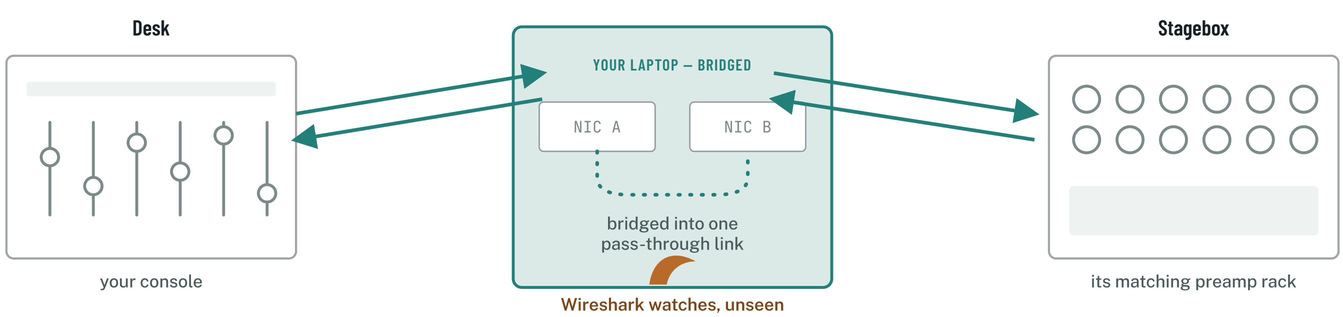
Watching Two Boxes Talk

A step-by-step guide to recording the private conversation between a mixing desk and its stagebox — using a laptop, two network cables, and one free piece of software. No specialist network gear required. This edition covers bridging on Windows.

NO TAP NEEDED | NO MIRROR SWITCH NEEDED | ~20 MINUTES | BENCH SETUP ONLY

THIS EDITION: **Windows** | macOS | Linux

DIAGRAM — THE IDEA IN ONE PICTURE 01



Every packet that would normally jump straight from desk to stagebox now physically passes through your laptop first — like fitting a clear length of pipe into a hose.

The whole trick: sit your laptop *in* the cable run, tell its two network ports to quietly pass everything through, and let Wireshark watch what goes by.

Every desk-and-stagebox pair from the same manufacturer has a private handshake — how they find each other, pair up, and agree on gain and phantom power — that's never been written down publicly. To build tools that can join that conversation from outside, we first have to record it happening between two devices that already trust each other.

> **Bench test, not showtime**
>
> Do this on a spare desk-and-stagebox pair, before or after a show — never on a live network mid-event. Inserting anything into a production Dante network, even briefly, isn't worth the risk.

§1 What you'll need

Nothing here is specialist. If you already run Dante gear, you likely own everything but the adapter cable — and that's a few pounds at any computer shop.

The desk

Your console.

REQUIRED

Its matching stagebox

Same brand and family as the desk — you want the pairing they already trust.

REQUIRED

Two Ethernet cables

The ordinary ones you'd use anyway.

REQUIRED

A laptop with two network ports

A built-in Ethernet port plus a cheap USB-to-Ethernet adapter covers almost any laptop.

REQUIRED

Wireshark, installed

Free at `wireshark.org` — Npcap, its packet driver, installs automatically alongside it.

REQUIRED

A mirroring switch or tap

Only if you already own one — see the alternatives near the end. Not required.

OPTIONAL

§2 The setup, step by step

Nine steps, done in order, from cabling to a saved file ready to hand over.

1 Wire it up

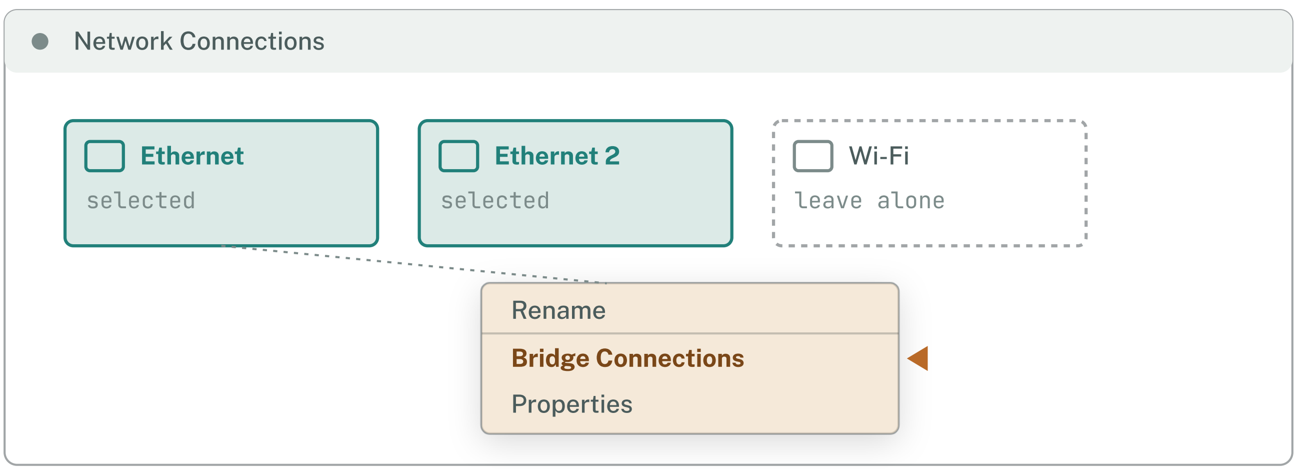
Run a cable from the desk's Ethernet port to **NIC A** on your laptop. Run a second cable from **NIC B** on your laptop to the stagebox's Ethernet port. That's it — no switch in between, your laptop *is* the connection now.

2 Bridge your two network ports

Tell your laptop to join the two ports into one silent pass-through, so nothing changes on the wire — just Wireshark gets to watch.

Open `Control Panel → Network Connections`, click your first adapter, `Ctrl`-click the second, right-click either one, and choose `Bridge Connections`. Windows creates a new `Network Bridge` adapter for you.

MOCKUP — WINDOWS NETWORK CONNECTIONS



Select both real adapters, right-click, choose *Bridge Connections*. Leave Wi-Fi out of it.

> Using a Mac or Linux box instead? There's a matching guide for each — the bridging step works differently, but everything else here is the same.

3 Open Wireshark and pick your interface

Look for `Network Bridge` in Wireshark's interface list — that's the one to capture on.

MOCKUP — WIRESHARK INTERFACE LIST

The Wireshark Network Analyzer

Network Bridge	← capture here
Ethernet	idle
Ethernet 2	idle
Wi-Fi	idle

Nothing on it yet — that's expected before anything's powered on.

If nothing shows up on `Network Bridge` once traffic is flowing, select `Ethernet` and `Ethernet 2` together instead and start both — Wireshark merges them into one capture automatically.

4

Start the capture first

Click the blue shark-fin `Start` button *before* powering anything on. This is what catches the very first discovery packets — the part we're actually missing today.

5

Power everything on

Turn the stagebox and desk on in whatever order you'd normally use, then wait about 30 seconds for them to find each other over the network.

6

Pair them, like normal

From the desk's own screen, connect to the stagebox exactly as you always would. No special settings — routine operation is exactly what we want recorded.

7

Wiggle a few things

Once, each: nudge a gain knob physically on the stagebox; nudge a different one from the desk's on-screen control; toggle phantom power on a channel; then disconnect the desk from the stagebox in software and reconnect it. That re-pairing moment is one of the most useful bits in the whole capture.

8

Stop and save

Hit the red `Stop` square, then `File → Save As`. Give it a name that says what it is — something like `ql1_tio1608_2026-07-14.pcapng` — and save.

9

Send it over, then tidy up

The `.pcapng` file on its own is enough — no need to trim or export anything yourself. Afterward, undo the bridge: right-click the `Network Bridge` adapter and choose `Remove from Bridge` on both ports, so your laptop goes back to normal.

§3 Already own a mirroring switch or a tap?

Neither is necessary — the laptop-bridge method above works fine on its own — but if you already have one of these, it's a perfectly good alternative to bridging.

DIAGRAM — THREE WAYS TO SEE THE SAME TRAFFIC

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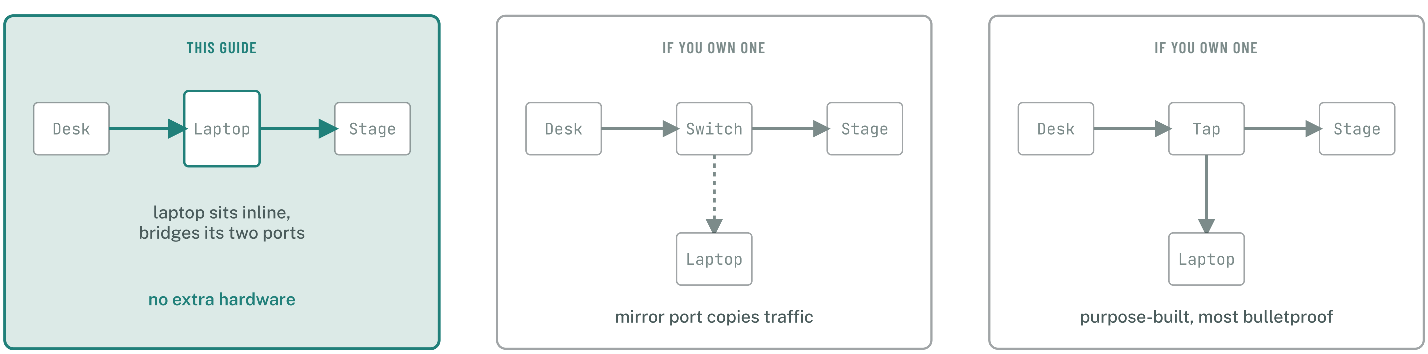

Managed switch with a mirror / SPAN port

Plug the desk, stagebox, and your laptop into the switch as normal, then configure one port to mirror the desk and stagebox's traffic onto your laptop's port. Capture directly on that single NIC — no bridging needed.

NEEDS ADMIN ACCESS TO THE SWITCH

Hardware network tap

A small dedicated box that sits inline between desk and stagebox, just like the laptop-bridge method, and quietly copies every packet out a third port to your laptop. The most bulletproof option, since it never routes the devices' whole conversation through a general-purpose computer.

EXTRA KIT TO BUY — NOT REQUIRED

§4 If something's not working

? I don't see "Bridge Connections" when I right-click

Make sure both adapters show as *Enabled* first, and that you've selected both together (Ctrl-click) before right-clicking — the option only appears with two selected at once.

? The Network Bridge interface shows no traffic

Capture on both individual physical adapters at the same time instead — Wireshark merges them into a single, correctly-ordered capture automatically.

? Windows Firewall seems to be blocking discovery

Temporarily allow the traffic through Windows Defender Firewall's private-network profile, or disable it just for the bench test — remember to turn it back on afterward.

? The desk and stagebox won't find each other through the laptop

Some laptops' Wi-Fi or antivirus network filtering can interfere with the multicast traffic discovery relies on. Turn Wi-Fi off entirely, or fall back to the mirror-port/tap method above.